

Duplicate BDM haloes

Why they survived, and the fix

DESI_MGx100 / DEGRACE nominal $z = 0.25$

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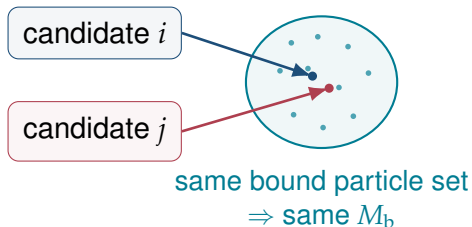


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Equal-mass candidates survived the overlap test

Two candidates can converge on one object



Recentring does not ensure a unique candidate.

Original removal rule

Remove i only if

$$d_{ij} < R_{\text{vir},j} \quad \text{and} \quad M_{b,i} < M_{b,j}.$$

For $M_{b,i} = M_{b,j}$, the mass test is false in both directions.

Both survive with separate halo IDs.

Duplicate rows inflate halo counts and add spurious close pairs.

Evidence: original BDM overlap rule and the same-particle-set synthetic reproduction. Diagram is schematic.

Merge each strict duplicate group deterministically

Require all four conditions

$$r_{ij} < 0.2 h^{-1} \text{Mpc}$$

$$M_{b,i} = M_{b,j}, \quad N_{b,i} = N_{b,j}$$

$$|\boldsymbol{v}_i - \boldsymbol{v}_j| < 5 \text{ km s}^{-1}$$

$$|\log_{10} M_{\text{tot},i} - \log_{10} M_{\text{tot},j}| < 0.005$$

Periodic separations; same strict criteria as Freyja.

One stable decision per group

1. Read unchanged measurements.
2. Connect all strict pairs.
3. Keep the **lowest original index** in each group.
4. Then apply the removal mask.

Same rule in the finder and cleaner; survivor properties are preserved.

Implementation: separate immutable host decision, followed by strict connected-component merging.

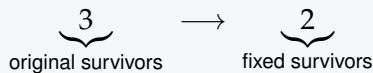
The failure is reproduced; retained values are preserved

Particle-level synthetic test

128 particles; candidates A, B, C .

A, B : identical sets of 64 IDs.

C : the other 64 particle IDs.



Passes with 1, 2 and 4 threads.

Retained properties are unchanged.

Reference catalogue: box 1

3,385,000 original rows

6,191 rows removed

3,378,809 rows retained

Same 6,191-row mask from the compiled Fortran rule and cleaner.

All 24 retained columns are **byte-identical**.

Production limit: no particle snapshots or membership files were found, so the reference replay is at catalogue level.

Evidence: source-based regressions, full finder build, and compiled box 1 policy replay.

All 740 catalogues are free of strict duplicate pairs

Nominal $z = 0.25$; fractions below use $\log_{10}[M_{\text{tot}}/(h^{-1}M_{\odot})] \geq 12.4$.

Sample	Catalogues	Removed fraction
Fiducial Λ CDM	100	0.309%
Λ CDM training	320	0.294%
fRn1 training	320	0.332%

Fiducials: removal rises to **0.924%** in the 13.4–13.6 mass bin.

All masses: 4.51 million rows removed from 2.50 billion.

740 cleaned copies and masks in the original row order are available.

Box 1 retains **1,041 ambiguous same-mass/count pairs**; their physical membership is unresolved.

Next: remeasure halo statistics, then retrain the emulators.

Evidence: completed 740-catalogue validation and mass-bin tables; RESULTS.md, 5 September 2026.